

Overview of BERT language model

Subject: BERT language model

1.

Masked language modeling (MLM): In this task, BERT ingests a sequence of words, where one word may be randomly changed ("masked"), and BERT tries to predict the original words that had been changed. For example, in the sentence "The cat sat on the [MASK]," BERT would need to predict "mat." This helps BERT learn bidirectional context, meaning it understands the relationships between words not just from left to right or right to left but from both directions at the same time. Next sentence prediction (NSP): In this task, BERT is trained to predict whether one sentence logically follows another. For example, given two sentences, "The cat sat on the mat" and "It was a sunny day", BERT has to decide if the second sentence is a valid continuation of the first one. This helps BERT understand relationships between sentences, which is important for tasks like question answering or document classification.

2.

with probability 80%, the chosen token is masked, resulting in "my1 dog2 is3 [MASK]4"; with probability 10%, the chosen token is replaced by a uniformly sampled random token, such as "happy", resulting in "my1 dog2 is3 happy4"; with probability 10%, nothing is done, resulting in "my1 dog2 is3 cute4". After processing the input text, the model's 4th output vector is passed to its decoder layer, which outputs a probability distribution over its 30,000-dimensional vocabulary space.

3.

Cost BERT was trained on the BookCorpus (800M words) and a filtered version of English Wikipedia (2,500M words) without lists, tables, and headers. Training BERTBASE on 4 cloud TPU (16 TPU chips total) took 4 days, at an estimated cost of 500 USD. Training BERTLARGE on 16 cloud TPU (64 TPU chips total) took 4 days.

4.

History BERT was originally published by Google researchers Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. The design has its origins from pre-training contextual representations, including semi-supervised sequence learning, generative pre-training, ELMo, and ULMFit. Unlike previous models, BERT is a deeply bidirectional, unsupervised language representation, pre-trained using only a plain text corpus. Context-free models such as word2vec or GloVe generate a single word embedding representation for each word in the vocabulary, whereas BERT takes into account the context for each occurrence of a given word. For instance, whereas the vector for "running" will have the same word2vec vector representation for both of its occurrences in the sentences "He is running a company" and "He is running a marathon", BERT will provide a contextualized embedding that will be different according to the sentence. On October 25, 2019, Google announced that they had started applying BERT models to English-language search queries on Google Search within the US. On December 9, 2019, it was reported that BERT had been adopted by Google Search for over 70 languages. In October 2020, almost every single English-based query was processed by a BERT model.

5.

The three attention matrices are added together element-wise, then passed through a softmax layer and multiplied by a projection matrix. Absolute position encoding is included in the final self-attention layer as additional input.

6.

Given two sentences, the model predicts if they appear sequentially in the training corpus, outputting either [IsNext] or [NotNext]. During training, the algorithm sometimes samples two sentences from a single continuous span in the training corpus, while at other times, it samples two sentences from two discontinuous spans. The first sentence starts with a special token, [CLS] (for "classify"). The two sentences are separated by another special token, [SEP] (for "separate"). After processing the two sentences, the final vector for the [CLS] token is passed to a linear layer for binary classification into [IsNext] and [NotNext]. For example:

7.

In masked language modeling, 15% of tokens would be randomly selected for masked-prediction task, and the training objective was to predict the masked token given its context. In more detail, the selected token is:

8.

Interpretation Language models like ELMo, GPT-2, and BERT, spawned the study of "BERTology", which attempts to interpret what is learned by these models. Their performance on these natural language understanding tasks are not yet well understood. Several research publications in 2018 and 2019 focused on investigating the relationship behind BERT's output as a result of carefully chosen input sequences, analysis of internal vector representations through probing classifiers, and the relationships represented by attention weights. The high performance of the BERT model could also be attributed to the fact that it is bidirectionally trained. This means that BERT, based on the Transformer model architecture, applies its self-attention mechanism to learn information from a text from the left and right side during training, and consequently gains a deep understanding of the context. For example, the word fine can have two different meanings depending on the context (I feel fine today, She has fine blond hair). BERT considers the words surrounding the target word fine from the left and right side. However it comes at a cost: due to encoder-only architecture lacking a decoder, BERT can't be prompted and can't generate text, while bidirectional models in general do not work effectively without the right side, thus being difficult to prompt. As an illustrative example, if one wishes to use BERT to continue a sentence fragment "Today, I went to", then naively one would mask out all the tokens as "Today, I went to [MASK] [MASK] [MASK] ... [MASK] ." where the number of [MASK] is the length of the sentence one wishes to extend to. However, this constitutes a dataset shift, as during training, BERT has never seen sentences with that many tokens masked out. Consequently, its performance degrades. More sophisticated techniques allow text generation, but at a high computational cost.

9.

Variants The BERT models were influential and inspired many variants. RoBERTa (2019) was an engineering improvement. It preserves BERT's architecture (slightly larger, at 355M parameters), but

improves its training, changing key hyperparameters, removing the next-sentence prediction task, and using much larger mini-batch sizes. XLM-RoBERTa (2019) was a multilingual RoBERTa model. It was one of the first works on multilingual language modeling at scale. DistilBERT (2019) distills BERTBASE to a model with just 60% of its parameters (66M), while preserving 95% of its benchmark scores. Similarly, TinyBERT (2019) is a distilled model with just 28% of its parameters. ALBERT (2019) used shared-parameter across layers, and experimented with independently varying the hidden size and the word-embedding layer's output size as two hyperparameters. They also replaced the next sentence prediction task with the sentence-order prediction (SOP) task, where the model must distinguish the correct order of two consecutive text segments from their reversed order. ELECTRA (2020) applied the idea of generative adversarial networks to the MLM task. Instead of masking out tokens, a small language model generates random plausible substitutions, and a larger network identify these replaced tokens. The small model aims to fool the large model. DeBERTa (2020) is a significant architectural variant, with disentangled attention. Its key idea is to treat the positional and token encodings separately throughout the attention mechanism. Instead of combining the positional encoding (x_{position}) and token encoding (x_{token}) into a single input vector ($x_{\text{input}} = x_{\text{position}} + x_{\text{token}}$), DeBERTa keeps them separate as a tuple: $(x_{\text{position}}, x_{\text{token}})$. Then, at each self-attention layer, DeBERTa computes three distinct attention matrices, rather than the single attention matrix used in BERT: